APPLIED PHYSICS FOR CSE STREAM(BPHYS102/202)

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https://braintube.site

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MODULE–4

ELECTRICAL PROPERTIES OF MATERIALS

ELECTRICAL CONDUCTIVITY IN METALS

1. Electrical Conductivity in Metals

Metals conduct electricity due to the presence of **free (conduction) electrons**.
When an external electric field $\vec{E}$is applied, electrons acquire a **drift velocity**, producing electric current.

2. Drift Velocity

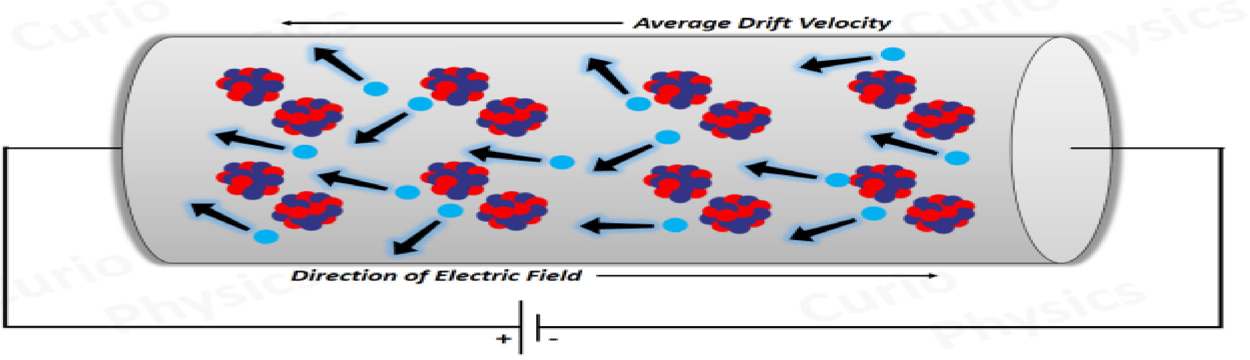
Definition

The **drift velocity** $v_d$is the average velocity with which free electrons drift opposite to the applied electric field.

$$\boxed{v_d = \frac{eE\tau}{m}}$$

where
e= electronic charge,
E= electric field,
τ= relaxation time,
m= mass of electron.

Drift of Electrons under Electric Field



3. Electrical Conductivity

Current density:

$$J = nqev_d$$

Using v_d,

$$J = \frac{ne^2\tau}{m}E$$

Hence, **electrical conductivity**:

$$\boxed{\sigma = \frac{ne^2\tau}{m}}$$

where
n= number of free electrons per unit volume.

4. Resistivity

Resistivity $\rho$is the inverse of conductivity:

$$\boxed{\rho = \frac{1}{\sigma}}$$

- Low resistivity → good conductor
- High resistivity → poor conductor

5. Mobility of Charge Carriers

Definition

Mobility $\mu$is the drift velocity per unit electric field.

$$\boxed{\mu = \frac{v_d}{E} = \frac{e\tau}{m}}$$

Higher mobility means electrons move more easily through the metal.

6. Concept of Phonon

A **phonon** is a quantum of lattice vibration in a solid.

- At higher temperatures, lattice vibrations increase
- Electron–phonon collisions increase
- Resistivity increases with temperatur

7. Matthiessen's Rule

Matthiessen's rule states that the **total resistivity** of a metal is the sum of:

- Resistivity due to lattice vibrations
- Resistivity due to impurities

$$\boxed{\rho = \rho_T + \rho_I}$$

where
ρ_T= temperature-dependent resistivity,
ρ_I= impurity resistivity.

8. Failures of Classical Free Electron Theory

Classical free electron theory fails to explain:

- Variation of electrical conductivity with temperature
- Electronic specific heat of metals
- Experimental values of conductivity
- Hall effect correctly

These limitations led to the **Quantum Free Electron Theory**.

9. Assumptions of Quantum Free Electron Theory

- Electrons obey **Pauli's exclusion principle**
- Electrons follow **Fermi–Dirac statistics**
- Only electrons near **Fermi energy** contribute to conduction
- Potential inside metal is constant

FERMI ENERGY, DENSITY OF STATES & FERMI FACTOR

10. Fermi Energy (E_F)

Definition

Fermi energy is the maximum energy of electrons in a metal at **absolute zero temperature (0 K)**.

Key points:

- At $T = 0$K, all states with $E \leq E_F$are occupied.
- States with $E > E_F$are empty.
- Only electrons near $E_F$participate in electrical conduction.

11. Density of States (DOS)

11.1 Definition

The **density of states** $g(E)$is the number of allowed electron states **per unit energy per unit volume** at energy E.

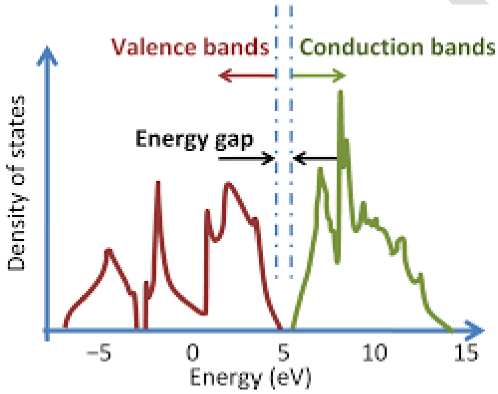
11.2 Expression (3D free electron gas)

$$\boxed{g(E) = \frac{1}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\sqrt{E}}$$

Observations:

- $g(E) \propto \sqrt{E}$
- Increases with energy
- Independent of temperature

Density of States vs Energy



12. Fermi–Dirac Distribution Function (Fermi Factor)

12.1 Definition

The **Fermi factor** $f(E)$gives the probability that an electron state of energy $E$is occupied at temperature T.

12.2 Expression

$$\boxed{f(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{kT}\right)}}$$

where
k= Boltzmann constant,
T= absolute temperature.

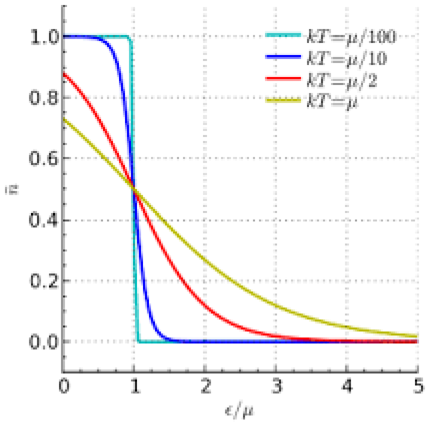
13. Properties of the Fermi Factor

- At $E = E_F$:

$$\boxed{f(E_F) = \frac{1}{2}\ \text{(for all } T\text{)}}$$

- At $T = 0$K:
    - $f(E) = 1$for $E < E_F$
    - $f(E) = 0$for $E > E_F$
- As Tincreases:
    - Distribution smoothens near E_F
    - Only a narrow energy range around $E_F$is affected

Fermi–Dirac Distribution vs Energy



14. Variation of Fermi Factor with Temperature and Energy

- **At 0 K**: sharp step function at E_F
- **At finite temperatures**: transition region of width $\sim kT$around E_F
- Occupation probability above $E_F$increases slightly with temperature

Implication:

- Electrical properties depend primarily on electrons near E_F

15. Number of Electrons in the Energy Range $E$to $E + dE$

The number of electrons available for conduction in the range $E$to $E + dE$is:

$$\boxed{N(E)\, dE = g(E)\, f(E)\, dE}$$

This product explains why:

- DOS alone is insufficient
- Fermi factor alone is insufficient
- **Both together** determine conduction

16. Numerical Problems (Typical)

Example 1

If $E = E_F$, find the occupation probability at any temperature.

$$f(E_F) = \frac{1}{1 + e^0} = \frac{1}{2}$$

Example 2

At $T = 0\text{K}$, determine $f(E)$ for $E < E_F$ and $E > E_F$.

- $E < E_F \Rightarrow f(E) = 1$
- $E > E_F \Rightarrow f(E) = 0$

SUPERCONDUCTIVITY

1. Introduction to Superconductivity

Superconductivity is a phenomenon in which certain materials exhibit **zero electrical resistivity** and **perfect diamagnetism** when cooled below a critical temperature T_c.

Key features:

- Zero resistivity
- Expulsion of magnetic field
- Occurs below a specific temperature

2. Temperature Dependence of Resistivity

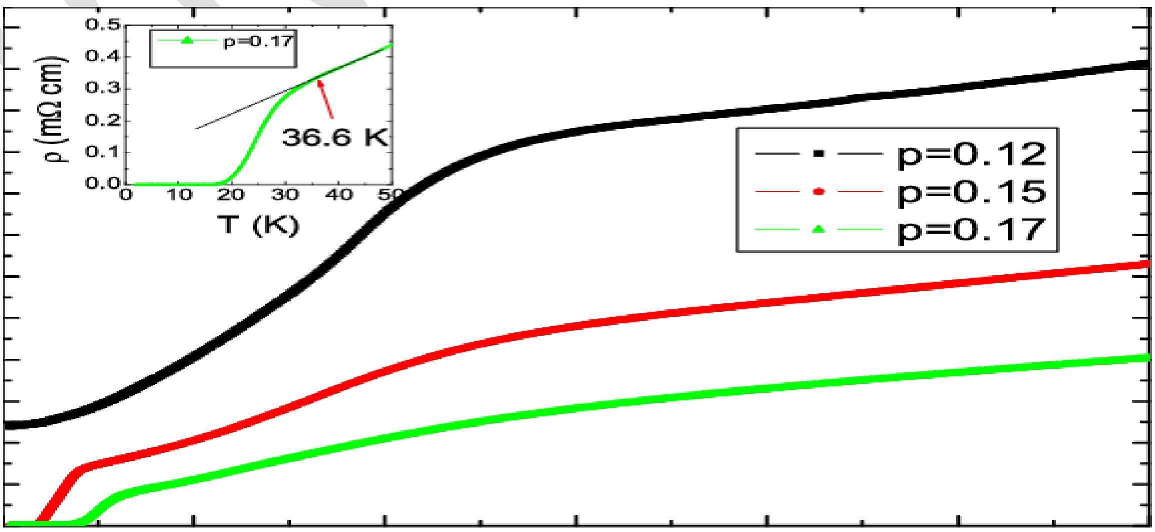
For a **normal conductor**:

- Resistivity decreases gradually with temperature

For a **superconductor**:

- Resistivity suddenly drops to **zero** at T_c

Resistivity vs Temperature Curve



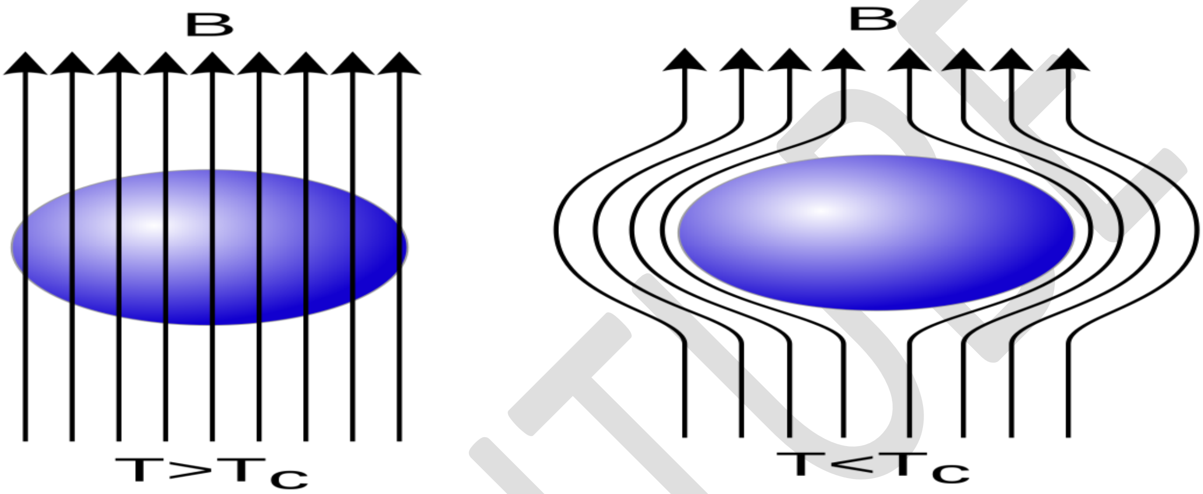
3. Meissner Effect

Definition

The **Meissner effect** is the complete expulsion of magnetic flux from the interior of a superconductor when it is cooled below T_c, even in the presence of an external magnetic field.

This shows that superconductivity is **not just perfect conductivity**, but a distinct thermodynamic state

Meissner Effect Diagram



4. Critical Magnetic Field (H_c)

The **critical magnetic field** is the maximum magnetic field that a superconductor can withstand at a given temperature without losing superconductivity.

If:

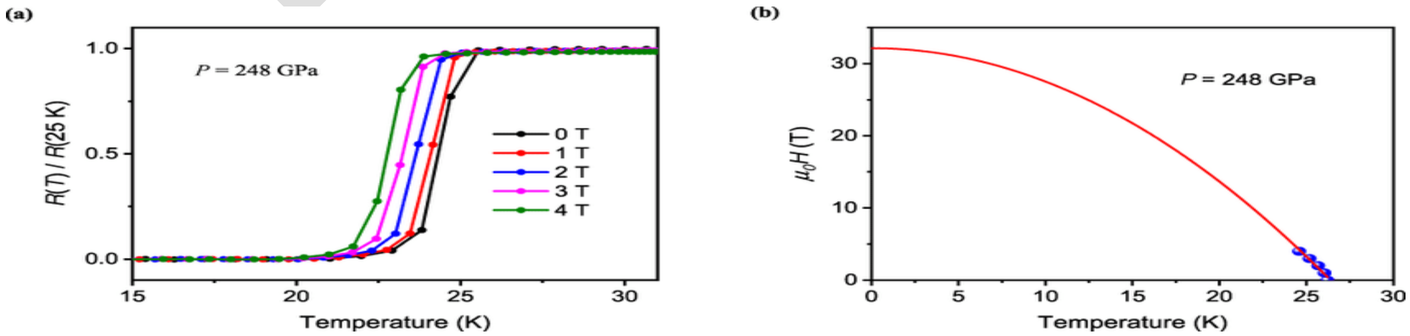
$$H > H_c \Rightarrow \text{material becomes normal}$$

Temperature Dependence of Critical Field

$$\boxed{H_c(T) = H_c(0)\left[1 - \left(\frac{T}{T_c}\right)^2\right]}$$

where
$H_c(0)$= critical field at absolute zero.

Critical Field vs Temperature Curve



5. Types of Superconductors

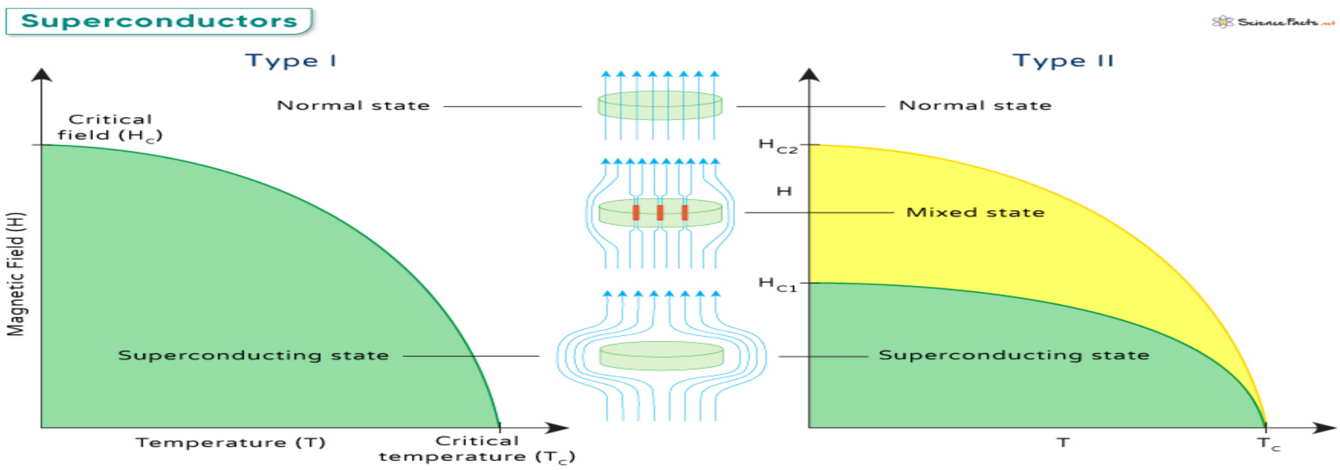
5.1 Type-I Superconductors

- Single critical field
- Complete Meissner effect
- Examples: Lead, Mercury, Aluminium

5.2 Type-II Superconductors

- Two critical fields H_{c1} and H_{c2}
- Partial penetration of magnetic field (mixed state)
- Examples: NbTi, Nb_3Sn

Type-I vs Type-II Comparison Diagram



6. BCS Theory (Qualitative)

According to **BCS theory**:

- Electrons near the Fermi level form **Cooper pairs**
- Pairing is due to electron–phonon interaction
- Cooper pairs move without scattering
- An **energy gap** forms at the Fermi surface

This explains:

- Zero resistivity
- Meissner effect

7. Quantum Tunnelling

Quantum tunnelling is the phenomenon where particles pass through a potential barrier even if their energy is less than the barrier height.

It plays a crucial role in:

- Josephson effect
- Superconducting devices

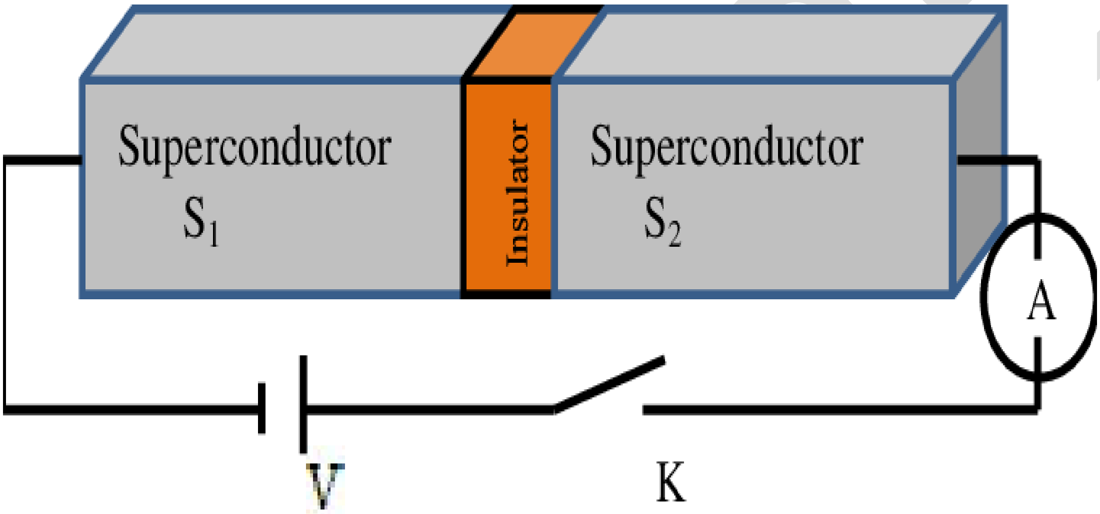
8. Josephson Junction

A **Josephson junction** consists of:

- Two superconductors
- Separated by a thin insulating barrier

Quantum tunnelling of Cooper pairs occurs across the barrier.

Josephson Junction Diagram



9. DC and RF SQUIDs (Qualitative)

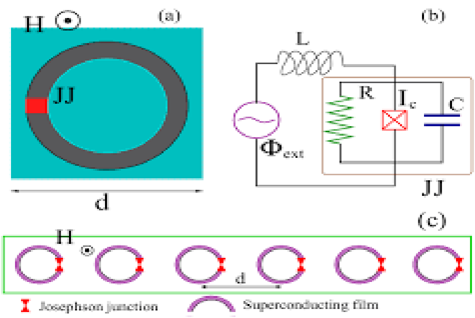
9.1 DC SQUID

- Consists of two Josephson junctions
- Extremely sensitive to magnetic fields

9.2 RF SQUID

- Contains a single Josephson junction
- Used for magnetic flux measurement

SQUID Devices Diagram (REQUIRED



10. Applications in Quantum Computing

Superconductors are used to create:

- **Charge qubits**
- **Flux qubits**
- **Phase qubits**

Advantages:

- Low energy loss
- High coherence time
- Used in modern quantum processors

11. Numerical Problems (Typical)

Example

A superconductor has $T_c = 8\,K$. Find the critical field at $T = 4\,K$, given $H_c(0) = 0.2\,T$.

$$H_c(4) = 0.2\left[1 - \left(\frac{4}{8}\right)^2\right]$$

$$H_c(4) = 0.15\,T$$